

State of SAP Exposure in 2025/2026



**Core Business
Application Security**

Executive Summary



- Internet scan conducted between **Nov 2025 – Jan 2026**
- **Thousands of SAP services exposed** on the public Internet
- **6744 SAPRouter instances** publicly reachable worldwide
- Misconfigurations allow **internal network information leakage**
- Critical vulnerabilities (e.g. **CVE-2025-31324**) still exposed
- Internet exposure of SAP infrastructure remains widespread and poses a global security risk
- SAP systems often store financial, supply chain and business-critical data

Responsible Research



- Only **non-intrusive internet scanning** was performed
- No **authentication attempts, brute force, or exploitation**
- Only **publicly accessible information** was analyzed
- Data gathered via **masscan, Shodan.io and nuclei validation**
- **Goal:** raise awareness of exposed SAP infrastructure

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Data Gathering Method

Notes



- Internet scan conducted **November 2025 – January 2026**
- For initial port identification, both free and commercial tools were used.
 - **Masscan** – great for services that do not run on “generic” ports (e.g. 80,8080,443,...)
 - **Shodan.io** – great to search for services based on responses like HTTP header, etc
- The Results of each service have been aggregated for research purposes (top 10 results)
- Only the IPv4 Address Space has been scanned

Notes



- Focus on **SAP services with high attack surface**
 - Large **attack surface / known exploits**
 - Services **typically not intended for internet exposure**
 - Components enabling **reconnaissance or lateral movement**
 - Frequently **misconfigured SAP services**

Process - General



Tool: masscan/shodan.io

Description:

Identify all IP addresses with a specific port open

Tool: awk/cut/tail

Description:

Format the results from masscan to „ip:port“ pairs

Tool: nuclei

Description:

Discover and validate if the listening service on the port is the correct service

Tool: nuclei

Description:

Scan and extract information from the service

Process - Evaluation



- IP addresses of identified services have been mapped using MaxMind GeoLite2City DB to get country details
- IP addresses of identified services have been mapped using MaxMind GeoLite2ASN DB to get the AS Number and Provider details
- The charts have been created using Microsoft Excel

List of Tools & Services used



- **SAP Attack Surface Discovery toolkit**

<https://github.com/SecuritySilverbacks/SAP-AttackSurfaceDiscovery>

- **Nuclei**

<https://github.com/projectdiscovery/nuclei>

- **Masscan**

<https://github.com/robertdavidgraham/masscan>

- **Shodan.io**

<https://www.shodan.io/>

Example Masscan Configuration



saprouter.conf

```
# Port 3299 (SAPRouter) scan
rate = 1000000.00
output-format = list
output-status = open
output-filename = 3299-results.txt
ports = 3299
range = 0.0.0.0/0
excludefile = exclude.txt
adapter-port = 3299
```

exclude.txt

```
10.0.0.0/8
172.16.0.0/12
192.168.0.0/16
127.0.0.0/8
100.64.0.0/10
```

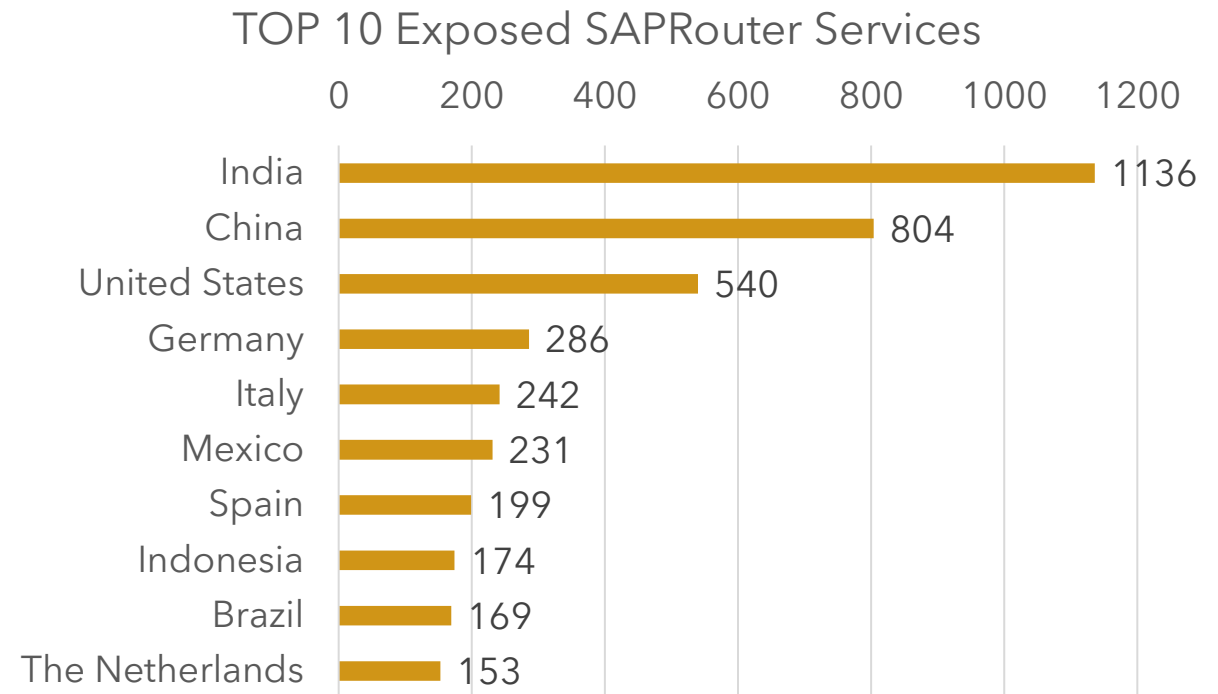


Results for SAPRouter

Results - SAPRouter



- **6744** SAPRouter services have been identified worldwide



Results - SAPRouter



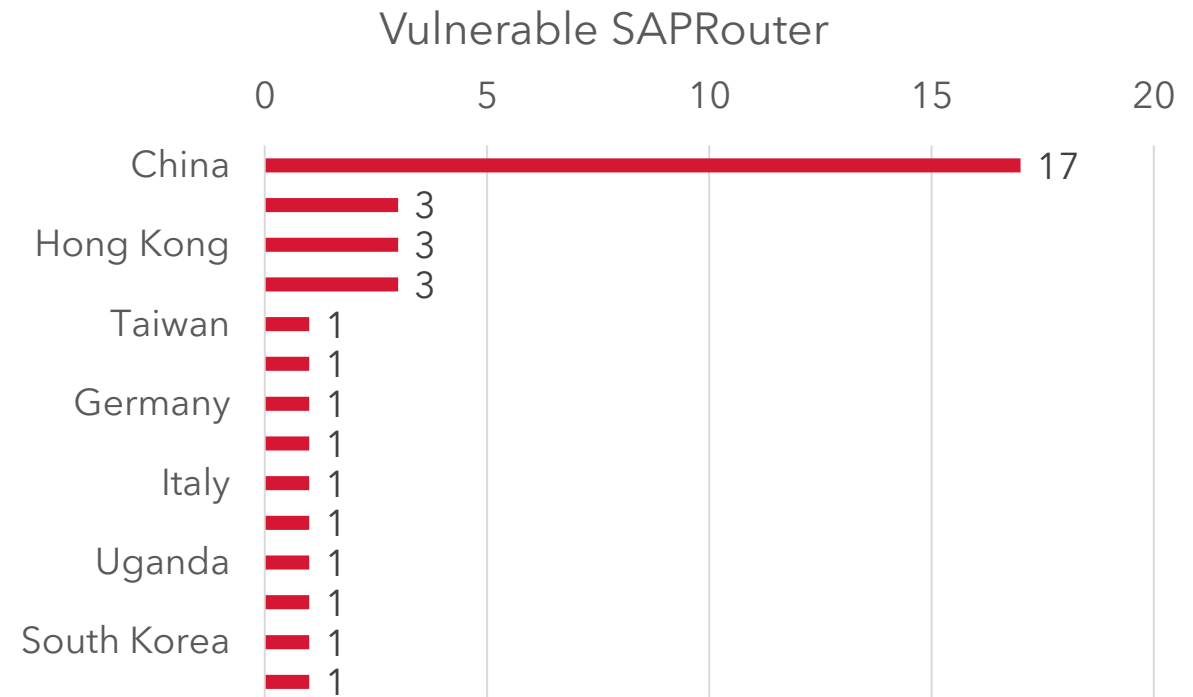
- **36** SAPRouter services have been identified to be not secured properly

- Allow to dump Connection table
- Leak internal IP addresses

[SAP] SAProuter Connection Table for [redacted]

Source	Destination	Service
192.168.[redacted]	192.168.[redacted]	3300
192.168.[redacted]	192.168.[redacted]	3300

- Exposure of Admin Interface



Results - SAPRouter

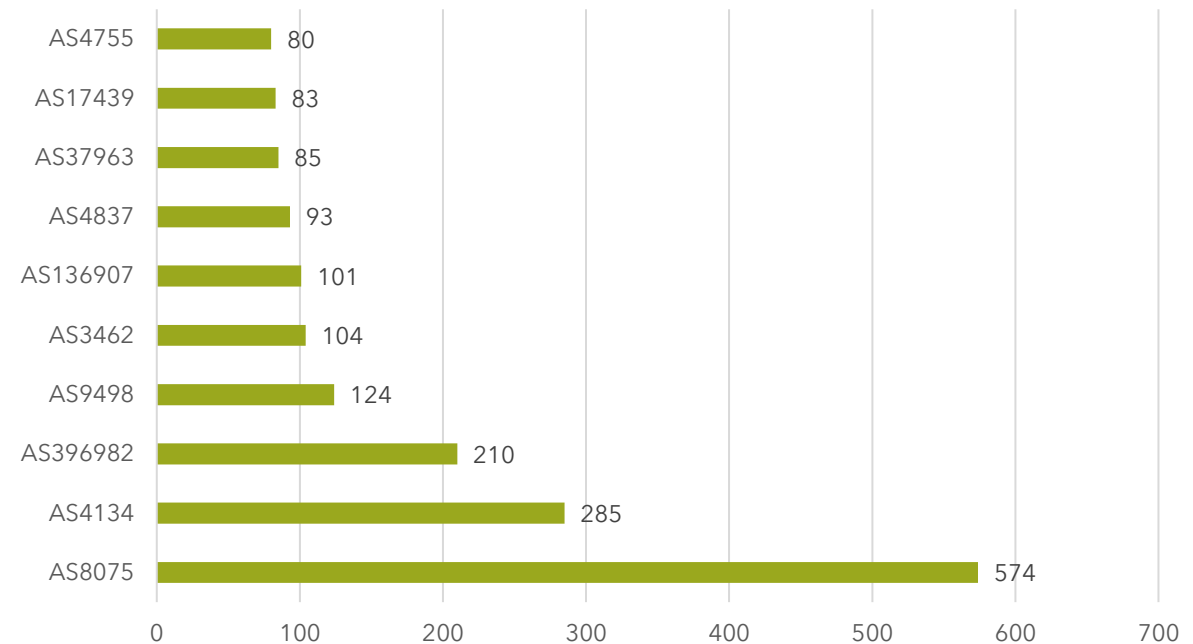


Top Cloud Providers Hosting This Service:

- Microsoft Azure
- Google Cloud
- Huawei Cloud

AS8075	MICROSOFT-CORP-MSN-AS-BLOCK
AS4134	CHINANET-BACKBONE
AS396982	GOOGLE-CLOUD-PLATFORM
AS9498	BBIL-AP (BHARTI Airtel Ltd.)
AS3462	HINET (Data Communication Business Group)
AS136907	HWLOUDS-AS-AP (HUAWEI CLOUDS)
AS4837	CHINA UNICOM China169 Backbone
AS37963	ALIBABA-CN-NET
AS17439	NTT COMMUNICATIONS INDIA NETWORK SERVICES PRIVATE LIMITED
AS4755	TATA Communications

TOP 10 Hosting ASN



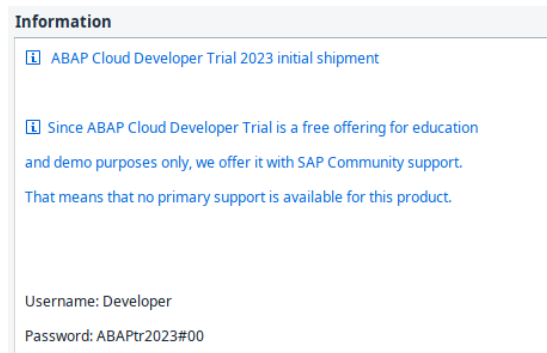


Results for SAP Dispatcher Service

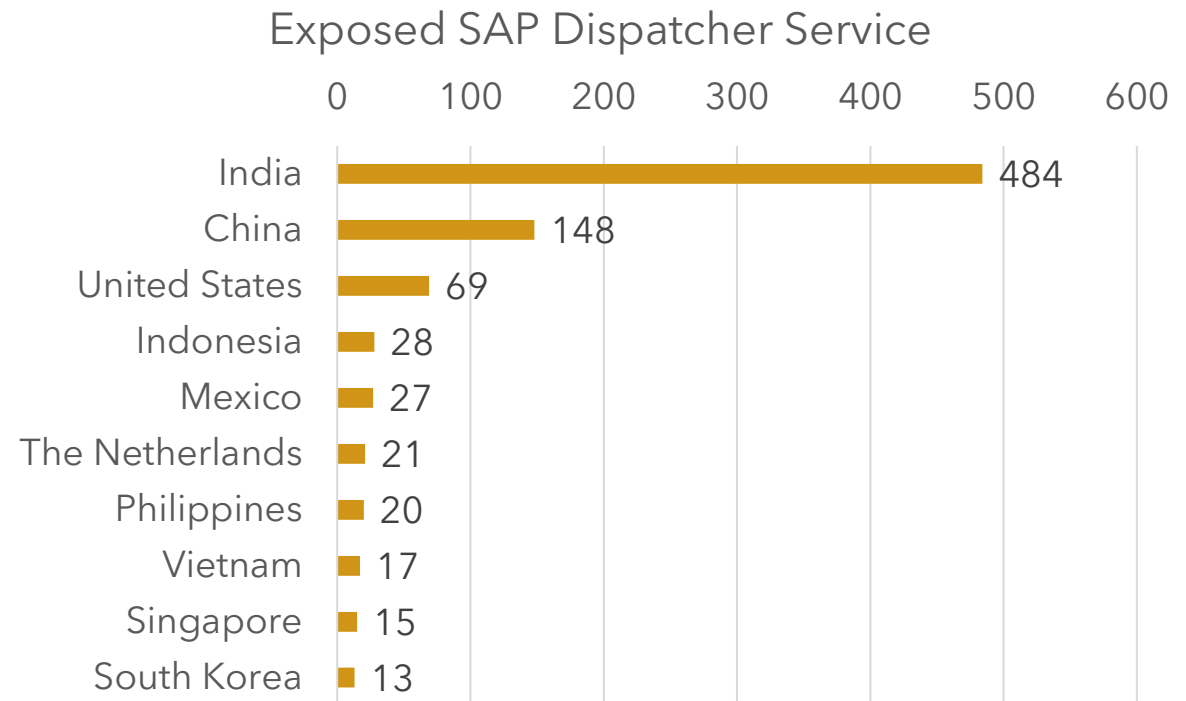
Results - SAP Dispatcher Service



- **914** SAP Dispatcher services have been identified
 - Allows reading the login screen information



- Allows brute forcing of usernames and passwords



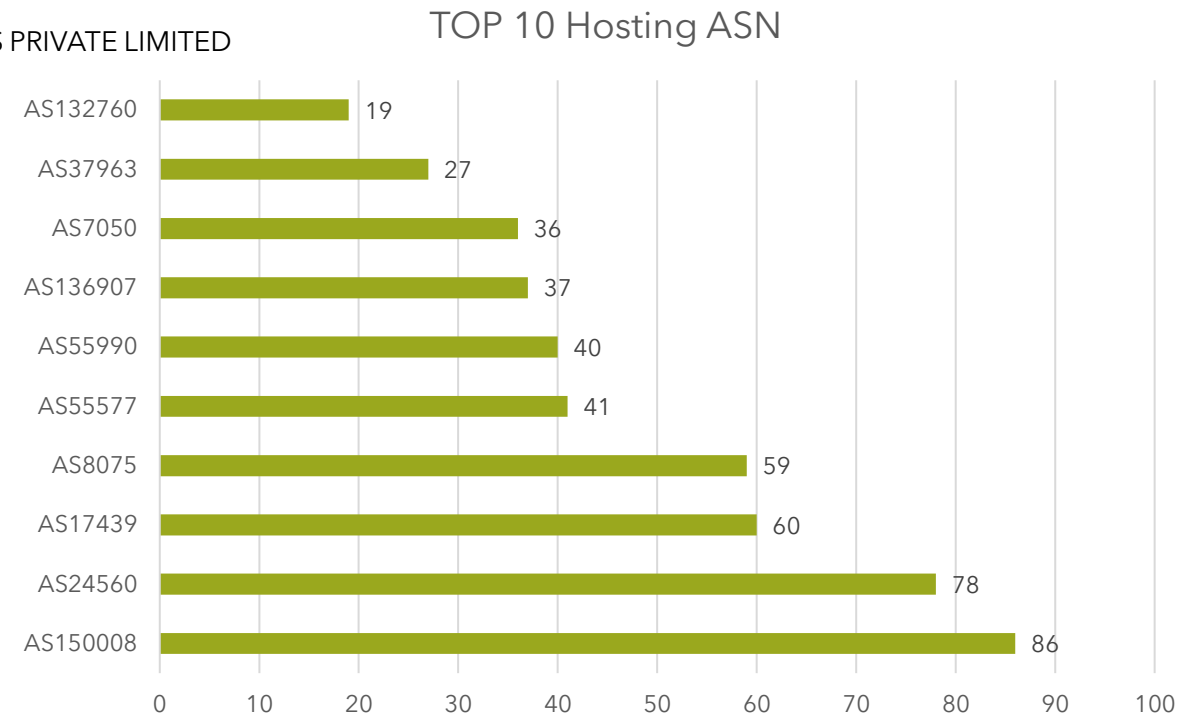
Results - SAP Dispatcher Service



AS150008	Pioneer Elabs Ltd.
AS24560	Bharti Airtel Ltd., Telemedia Services
AS17439	NTT COMMUNICATIONS INDIA NETWORK SERVICES PRIVATE LIMITED
AS8075	Microsoft Corporation
AS55577	Atria Convergence Technologies Ltd.
AS55990	Huawei Cloud Service data center
AS136907	HUAWEI CLOUDS
AS7050	University of Wisconsin - Milwaukee
AS37963	Hangzhou Alibaba Advertising Co.,Ltd.
AS132760	Progression Infonet Pvt Ltd

Top Cloud Providers Hosting This Service:

- Microsoft Azure
- Huawei Cloud
- Alibaba Cloud



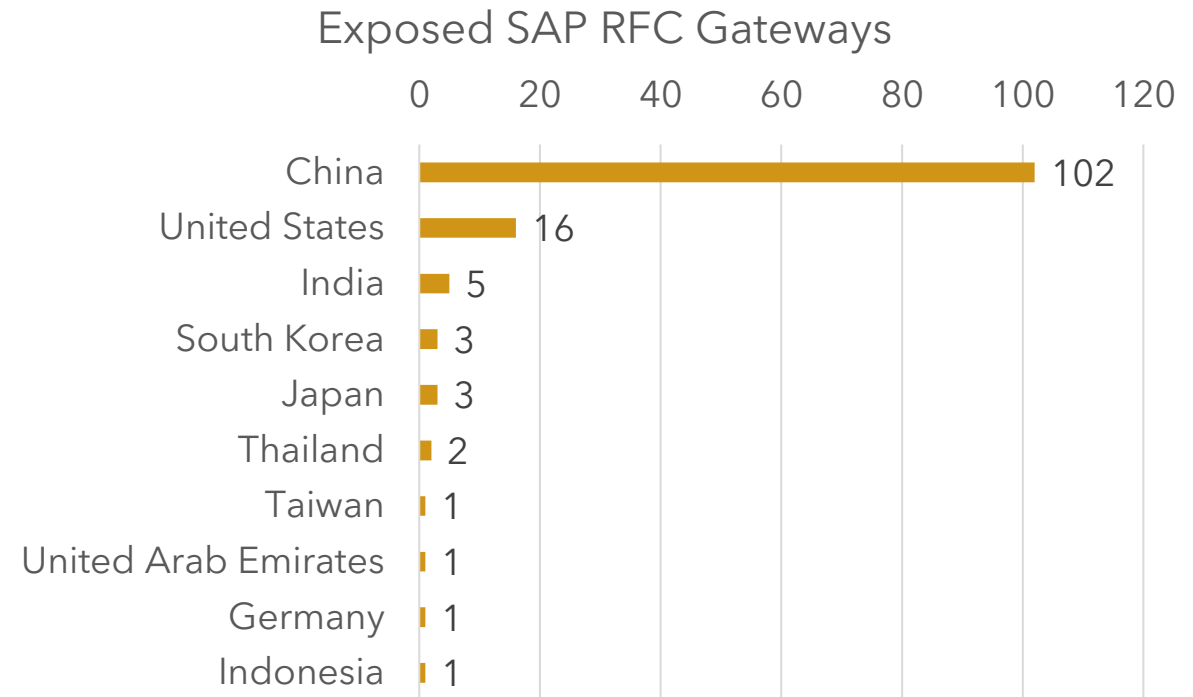


Results for SAP RFC Gateway

Results - SAP RFC Gateway



- **135** SAP RFC Gateway services have been identified
- **89** RFC Gateway Services were listening on the same IP address



Results - SAP RFC Gateway

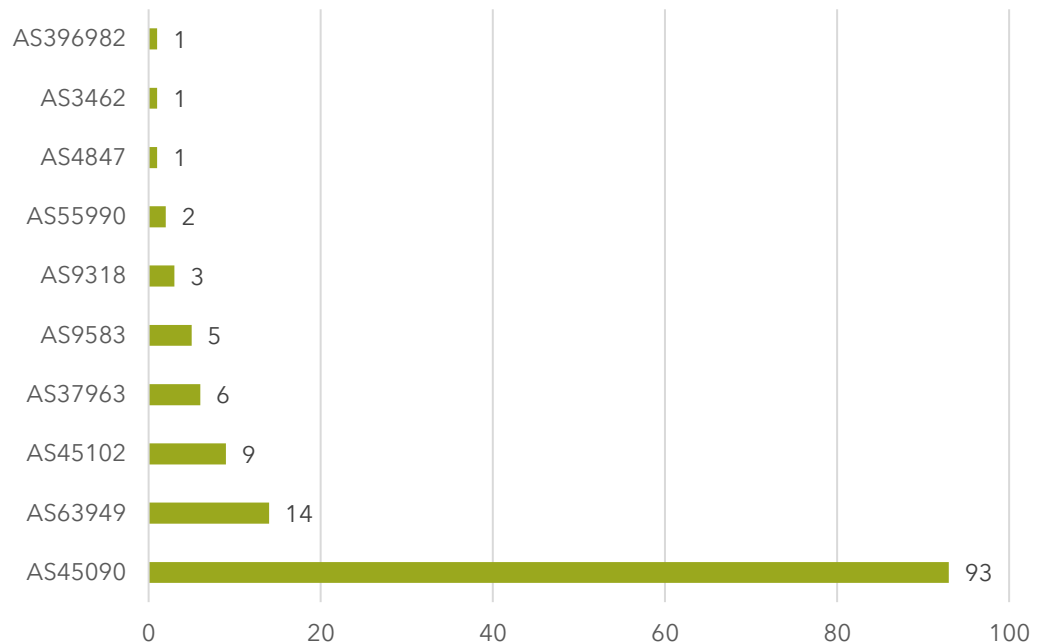


AS45090	Shenzhen Tencent Computer Systems Company Limited
AS63949	Akamai Connected Cloud (Linode)
AS45102	Alibaba US Technology Co., Ltd.
AS37963	Hangzhou Alibaba Advertising Co.,Ltd.
AS9583	Sify Limited
AS9318	SK Broadband Co Ltd
AS55990	Huawei Cloud Service data center
AS4847	China Networks Inter-Exchange
AS3462	Data Communication Business Group
AS396982	Google LLC

Top Cloud Providers Hosting This Service:

- Alibaba Cloud
- Linode
- Huawei Cloud

TOP 10 Hosting ASN



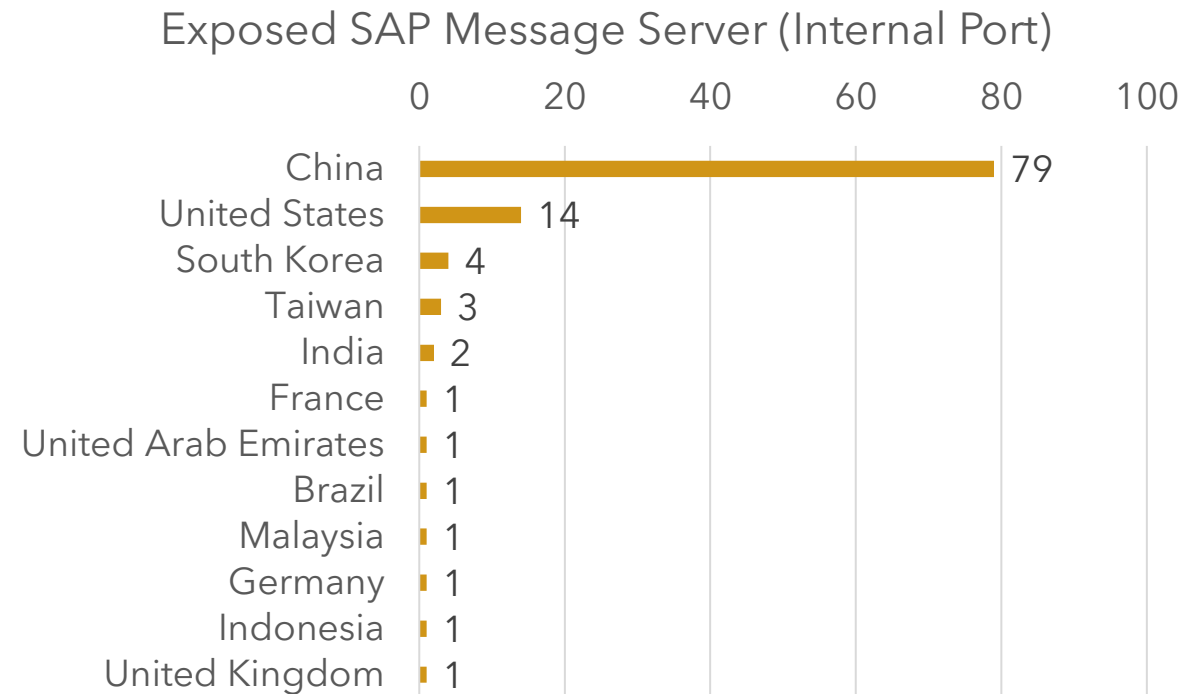


Results for SAP Message Server (internal)

Results - SAP Message Server (internal)



- **109** Internal SAP Message Server services have been identified
- **73** Internal SAP Message Server were listening on the same IP address



Results - SAP Message Server (internal)

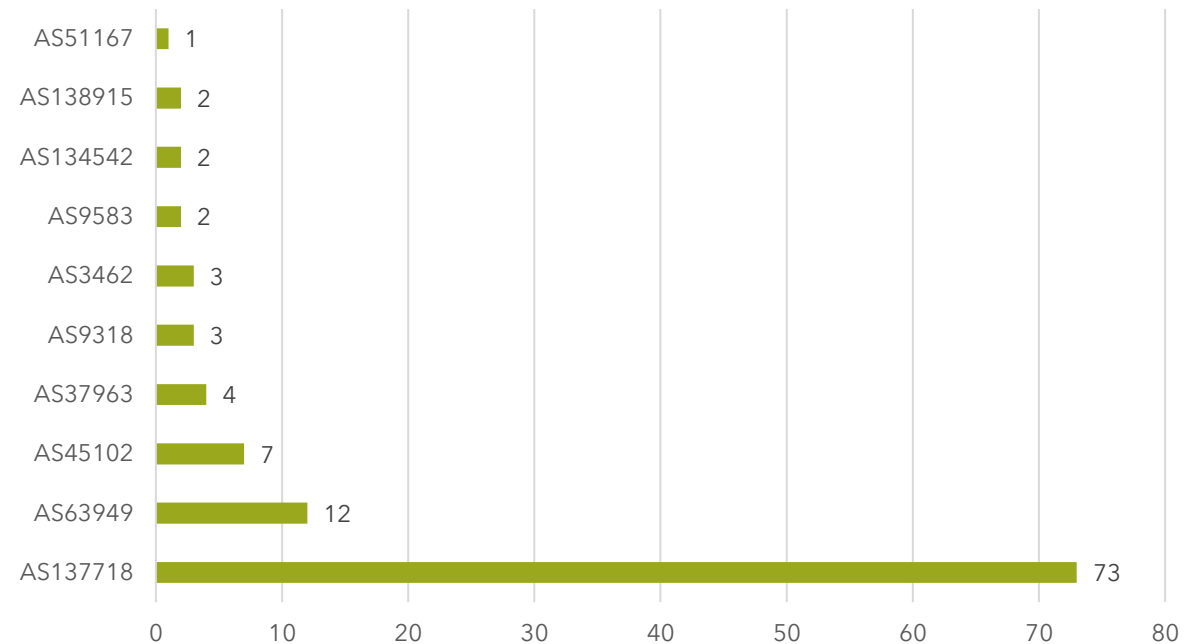


Top Cloud Providers Hosting This Service:

- Linode
- Alibaba Cloud
- Google Cloud

AS137718	Beijing Volcano Engine Technology Co., Ltd.
AS63949	Akamai Connected Cloud (Linode)
AS45102	Alibaba US Technology Co., Ltd.
AS37963	Hangzhou Alibaba Advertising Co.,Ltd
AS9318	SK Broadband Co Ltd
AS3462	Data Communication Business Group
AS9583	Sify Limited
AS134542	China Unicom IP network
AS138915	Kaopu Cloud HK Limited
AS51167	Contabo GmbH

TOP 10 Hosting ASN



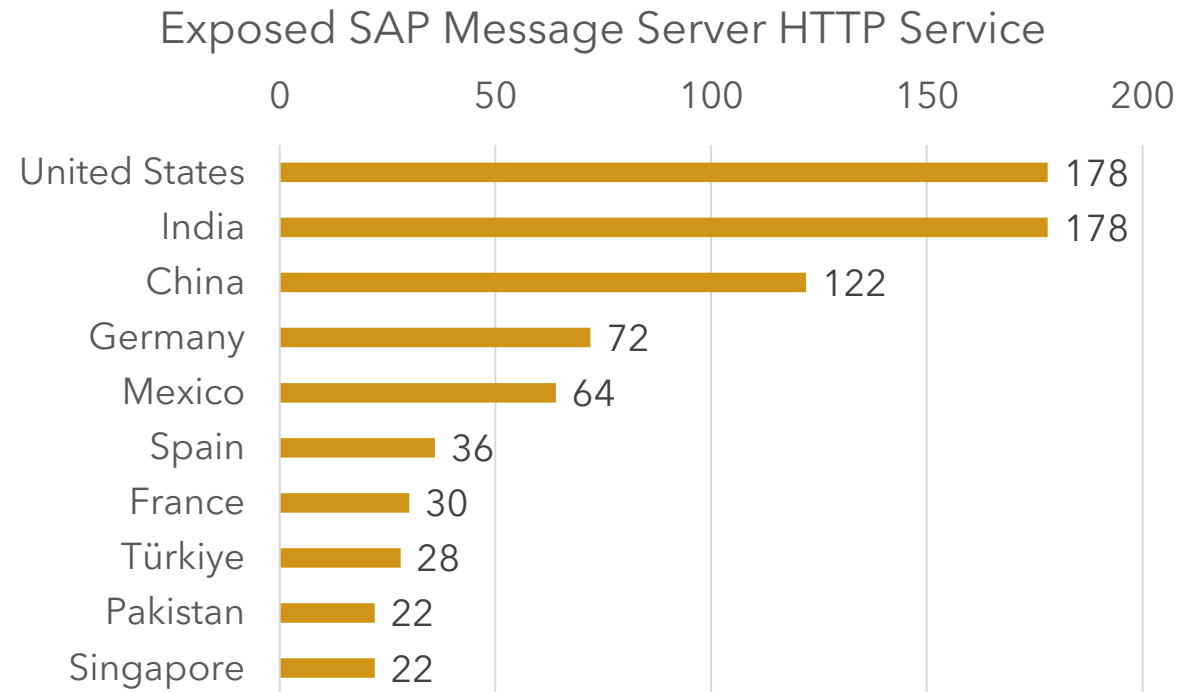


Results for SAP Message Server (HTTP)

Results - SAP Message Server (HTTP)



- **1032** SAP Message Server (HTTP) services have been identified
 - Allows to enumerate SAP profile parameters
 - Allows to enumerate internal IP addresses, hostnames and existing servers



The initial data has been gathered using Shodan.io

Results - SAP Message Server (HTTP)

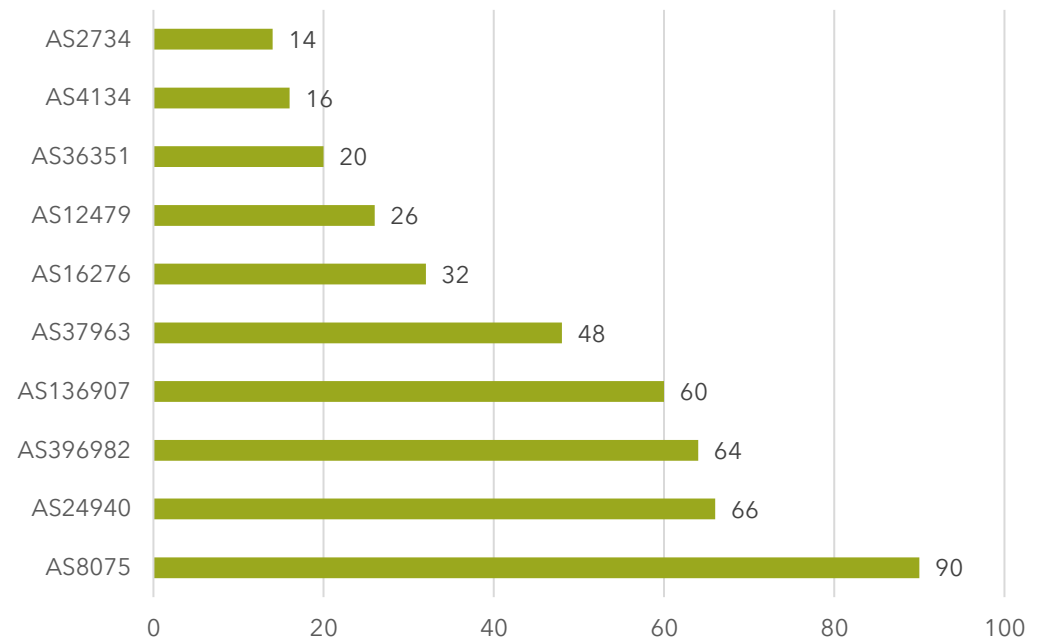


Top Cloud Providers Hosting This Service:

- Microsoft Azure
- Hetzner
- Google Cloud

AS8075	Microsoft Corporation
AS24940	Hetzner Online GmbH
AS396982	Google LLC
AS136907	HUAWEI CLOUDS
AS37963	Hangzhou Alibaba Advertising Co.,Ltd.
AS16276	OVH SAS
AS12479	Orange Espagne SA
AS36351	IBM Cloud
AS4134	Chinanet
AS2734	CoreSite

TOP 10 Hosting ASN



The initial data has been gathered using Shodan.io

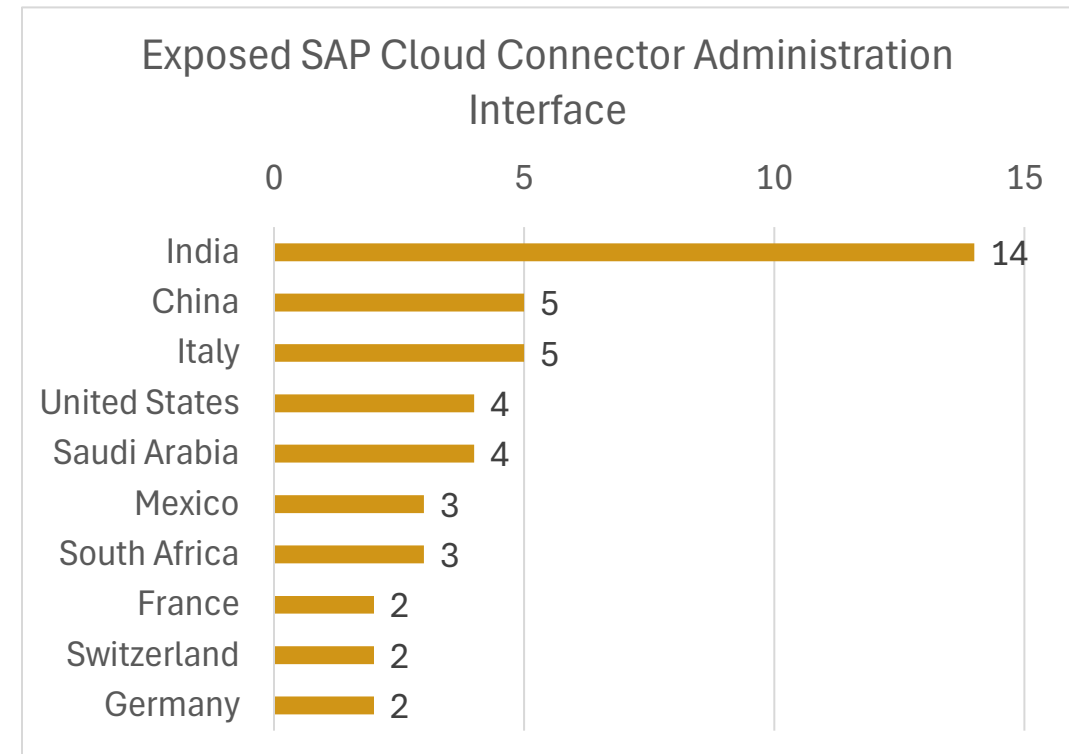


Results for SAP Cloud Connector

Results - SAP Cloud Connector



- **61** SAP Cloud Connector services have been identified
 - All Instances had their Administration Interface exposed



The initial data has been gathered using Shodan.io

Results - SAP Cloud Connector

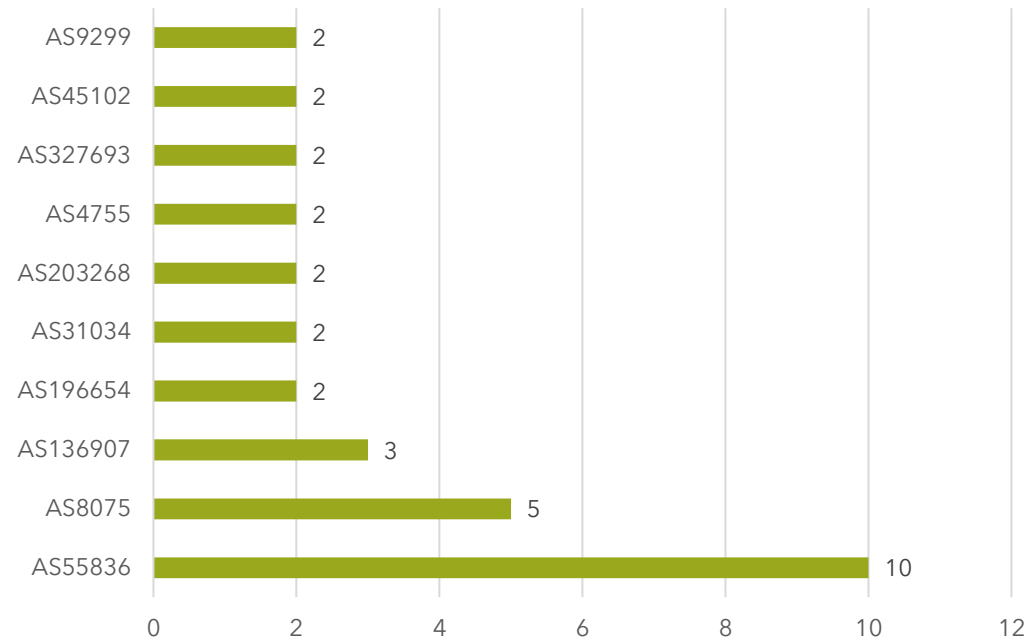


Top Cloud Providers Hosting This Service:

- Microsoft Azure
- Hetzner
- Google Cloud

AS55836	Reliance Jio Infocomm Limited
AS8075	Microsoft Corporation
AS136907	HUAWEI CLOUDS
AS196654	Deda Tech SRL
AS31034	Aruba S.p.A.
AS203268	Digital Technology Co.Ltd.
AS4755	TATA Communications
AS327693	ECHO-SP
AS45102	Alibaba US Technology Co., Ltd.
AS9299	Philippine Long Distance Telephone Company

Top 10 Hosting ASN



The initial data has been gathered using Shodan.io

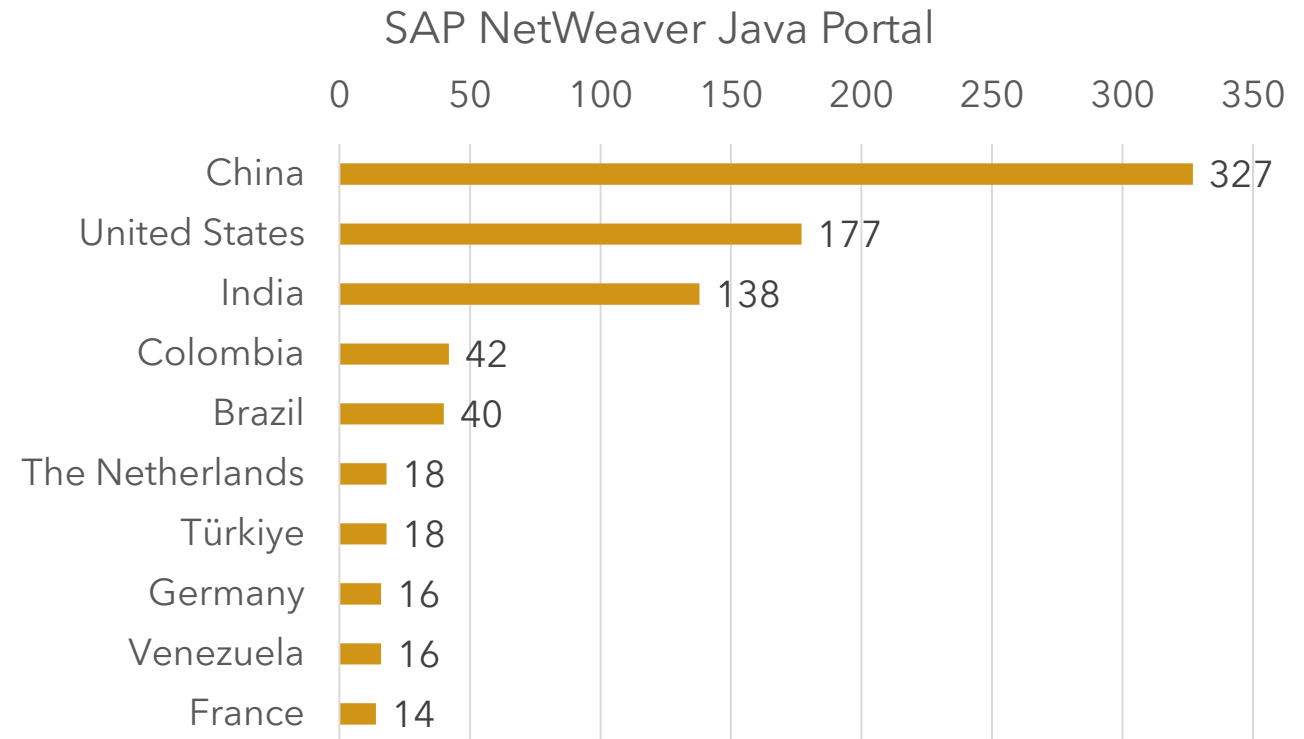


Results for SAP NetWeaver Java (HTTP)

Results - SAP NetWeaver Java Portals



- **974** SAP NetWeaver Java Portal services have been identified

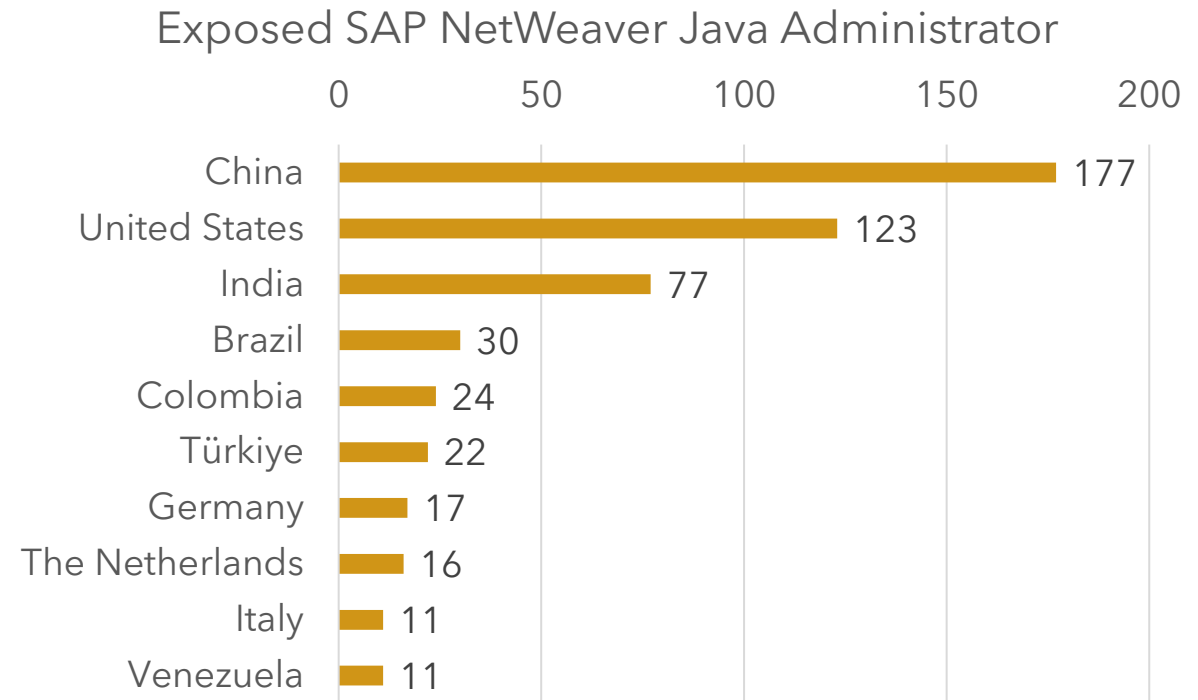


The initial data has been gathered using Shodan.io

Results - SAP NetWeaver Java NWA



- **647** SAP NetWeaver Java Administrator services have been identified

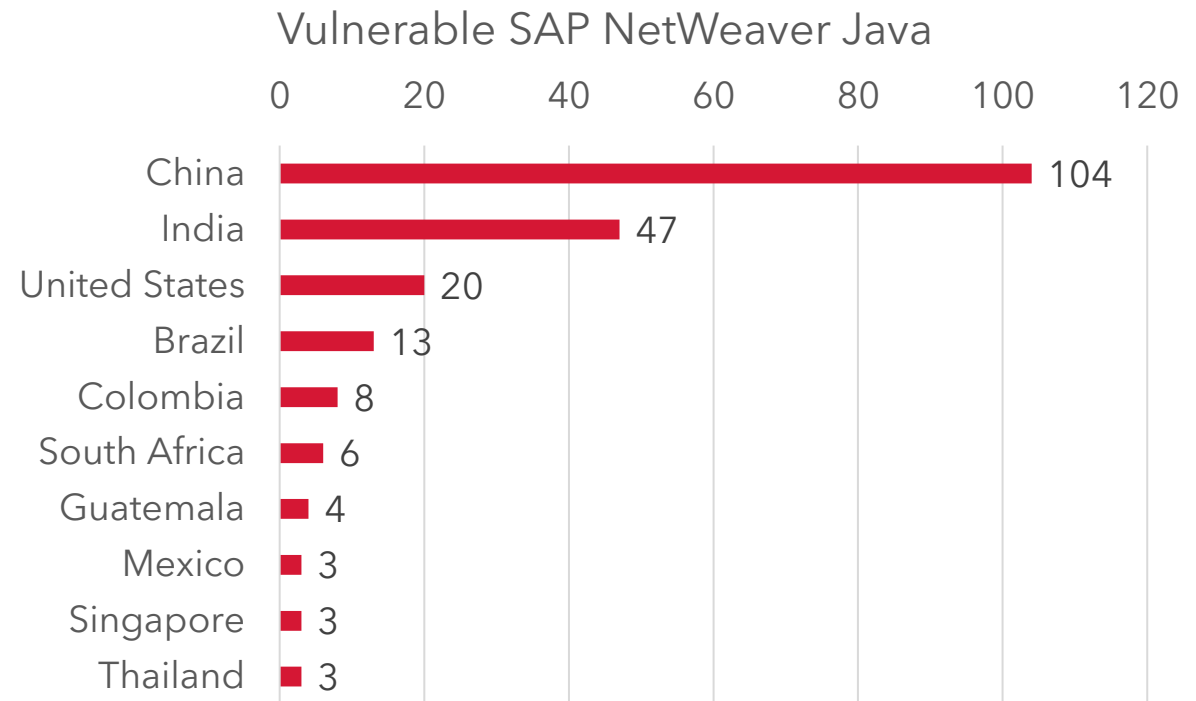


The initial data has been gathered using Shodan.io

Results - SAP NetWeaver Java Vuln



- **233** vulnerable SAP NetWeaver Java have been identified
- **Vulnerability:** CVE-2025-31324
 - Remote Code Execution
 - CVSS: 10.0



The initial data has been gathered using Shodan.io

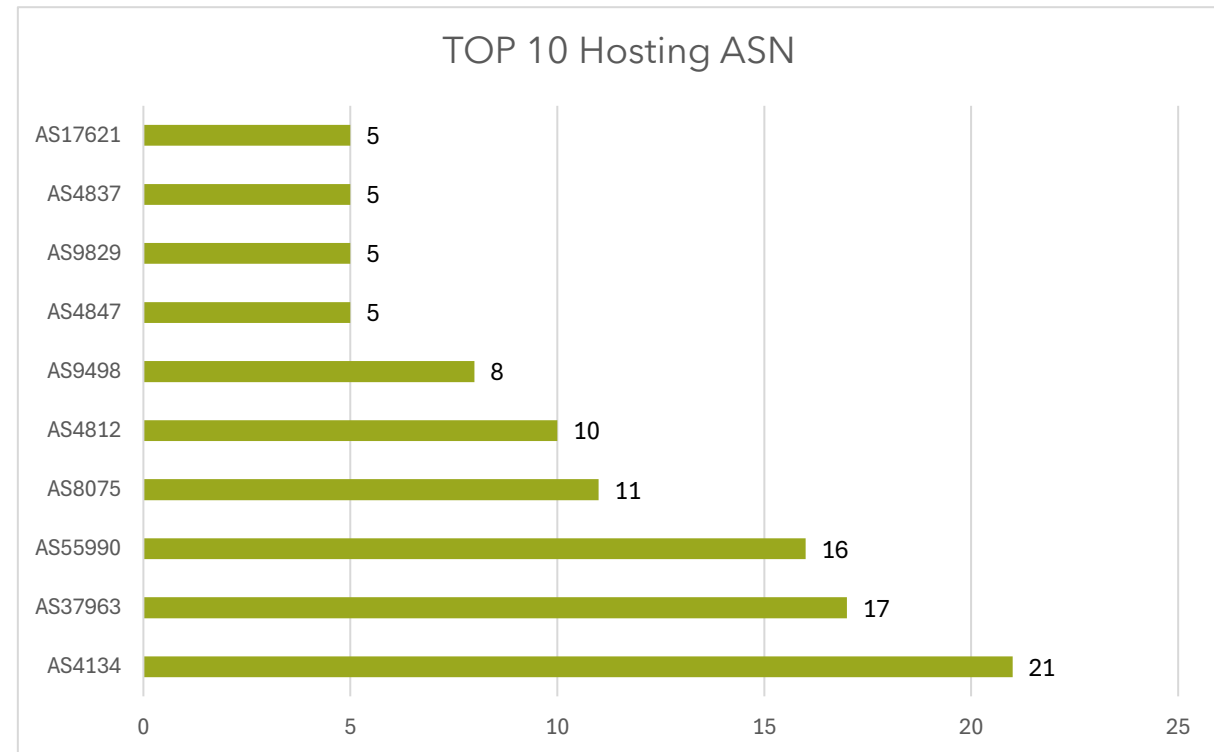
Results - SAP NetWeaver Java Vuln



Top Cloud Providers Hosting This Service:

- Alibaba Cloud
- Huawei Cloud
- Microsoft Azure

AS4134	Chinanet
AS37963	Hangzhou Alibaba Advertising Co.,Ltd.
AS55990	Huawei Cloud Service data center
AS8075	Microsoft Corporation
AS4812	China Telecom Group
AS9498	BHARTI Airtel Ltd.
AS4847	China Networks Inter-Exchange
AS9829	National Internet Backbone
AS4837	CHINA UNICOM China169 Backbone
AS17621	China Unicom Shanghai network



The initial data has been gathered using Shodan.io

Conclusion

- Thousands of SAP services remain **publicly exposed**
- Misconfigurations still expose **internal system information**
- **Critical vulnerabilities remain reachable**
- Organizations should regularly **audit external SAP exposure**



Check out the OWASP Core Business Application Security Project!

